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REPORT PROSODY (S.RUVOLETTO)

Github repository is available with all the documentation and process:
https://github.com/MaeDj/prosody_M1_2ndSemester_Ru

Research Questions

The nativity of the speaker in the French language was chosen as the main focus of exploration for this work.

The main hypothesis (H0) was to determine whether the peption of French vowels is the same for native and non-native speakers of French.

To achieve this goal, two different features were considered:

- Accuracy (How many mistakes the participant makes across the test)

- Reaction time (calculated in milliseconds)

Another avenue of exploration was also considered: the robustness of this first hypothesis and the influence of other factors on it.

For that, we studied:

- **Device** (headphones or speakers)

- **French level** (lower level → *below the median French level of our test group* or higher level)

- **French listening rate** (lower level or higher level)

- **French years of practice** (fewer →*below the median of the group* or more)

- **gender** (male or female, *due to the small amount of data, some approximations were made about the definition of gender. People with a non-binary gender were classified into the "male" group by default.*)

- **reaction time** (lower score or higher score)

- **nasality** (

nasal: ("bant": "/bɑ̃/", "dant": "/dɑ̃/", "gant": "/gɑ̃/", "kant": "/kɑ̃/", "pant": "/pɑ̃/", "tant": "/tɑ̃/", "bont": "/bɔ̃/", "dont": "/dɔ̃/", "gont": "/gɔ̃/", "kont": "/kɔ̃/", "pont": "/pɔ̃/", "tont": "/tɔ̃/", "bunt": "/bœ̃/", "dunt": "/dœ̃/", "gunt": "/gœ̃/", "kunt": "/kœ̃/", "punt": "/pœ̃/", "tunt": "/tœ̃/"),

not nasal/oral: ("bat": "/ba/", "dat": "/da/", "gat": "/ga/", "kat": "/ka/", "pat": "/pa/", "tat": "/ta/", "bet": "/bɛ/", "det": "/dɛ/", "get": "/gɛ/", "ket": "/kɛ/", "pet": "/pɛ/", "tet": "/tɛ/", "bot": "/bɔ/", "dot": "/dɔ/", "got": "/gɔ/", "kot": "/kɔ/", "pot": "/pɔ/", "tot": "/tɔ/")

- **Sex of the speaker** (male or female)

Participants

This work is based, for the main hypothesis, on two groups:

- Native speakers of French

- Non-native speakers of French from diverse countries (Portugal 1, Romania 1, Pulaar 1, English 3, Slovene 1, Korean 1, Italian 3, Arabic 1, German 1, Turkish 1, Slovak 1)

Two features were then measured to compare these two groups: accuracy and reaction time.

For the robustness hypothesis, we worked with 8 modalities and 22 sub-groups in addition to the 2 global groups with all native speakers on one side and all non-native speakers on the other:

- Native vs. non-native people who used headphones

- Native vs non-native people who used speakers

- All native vs. non-native people with a lower level in French

- All native vs. non-native people with a higher level in French

- All native vs. non-native people with a lower French listening rate

- All native vs. non-native people with a higher French listening rate

- All native vs. non-native people with fewer years of French practice

- All native vs. non-native people with more years of French practice

- Native vs. non-native people who identify as male or non-binary

- Native vs. non-native people who identify as female

- Native vs non-native people with a lower reaction time

- Native vs. non-native people with a higher reaction time

- Native vs. non-native people responding to nasal vowels

- Native vs. non-native people responding to non-nasal vowels

- Native vs non-native people answering to audio recorded by a female voice

- Native vs. non-native people responding to audio recorded by a male voice

“All native…” group is used when the original sub-group had an insufficient number of participants, or when it does not exist — as is the case for “Native speakers with lower level in French”. This group does not exist — since by definition, a native French speaker’s French level is always “Native.” This sub-group is used 6 times so we have:

2 global groups+((**2** features × **8** modalities*__2__ modality sub-groups)-**6** times the global group is used)

= **24** groups studied in total across the main hypothesis and robustness testing of the main hypothesis.

The dataset consists of 17 native French speakers out of 33 participants. For non-native French speakers, the dataset shows great diversity, with 12 different native languages represented. Regarding the age of the participants, all of them fall within the same range (20–30 years old), which is why age was not selected as a modality to test the robustness of the main hypothesis.

Concerning the gender modality, this dataset does not present a clear distinction between sex and gender. Moreover, given the size of the dataset, it was not possible to create a separate category for non-binary gender. That is why non-binary gender was classified into the “male” category by default. This choice is highly debatable but is solely due to the time constraints of this work and the low number of participants. It would need to be reconsidered in any professional or public research context.

The values for years of French practice are also not well standardized. Some native participants left the field blank, while others entered their age. To address this issue, the global native group was used by default for all comparisons with the non-native group having more or fewer years of French practice.

Tasks

The task we designed is a discrimination task. Participants are asked to listen to pairs of pronounced made-up syllables and decide whether they are hearing the same word or two different words. We measure their accuracy and reaction time.

In total, there are 144 pairs that a participant goes through, for an average duration of approximately 5–10 minutes. The structure in the software is designed as follows:

- Structure of each trial: The order of pairs is randomized to prevent priming from having a statistically significant effect. A random delay is also added between the words of each pair to counteract guesses based on the rhythm of our enunciations.
- Progress Tracking: Participants can see how far they have progressed in completing the study, in order to keep them focused.
- Practice: Before starting the main test, participants are given 1 practice trial to ensure they understand the interface.
- Availability: Since we hosted the experiment on a website that we sent to friends and family, they can decide when they have the time to complete the study and not have to stop half-way through.

Material

We recorded a total of 36 stimuli, each based on the following list: `STIMULI = [("bant", "bɑ̃"), ("bint", "bɛ̃"), ("bont", "bɔ̃"), ("pant", "pɑ̃"), ("pint", "pɛ̃"), ("pont", "pɔ̃"), ("tant", "tɑ̃"), ("tint", "tɛ̃"), ("tont", "tɔ̃"), ("dant", "dɑ̃"), ("dint", "dɛ̃"), ("dont", "dɔ̃"), ("kant", "kɑ̃"), ("kint", "kɛ̃"), ("kont", "kɔ̃"), ("gant", "gɑ̃"), ("gint", "gɛ̃"), ("gont", "gɔ̃"), ("bat", "ba"), ("bet", "be"), ("bot", "bo"), ("pat", "pa"), ("pet", "pe"), ("pot", "po"), ("tat", "ta"), ("tet", "te"), ("tot", "to"), ("dat", "da"), ("det", "de"), ("dot", "do"), ("kat", "ka"), ("ket", "ke"), ("kot", "ko"), ("gat", "ga"), ("get", "ge"), ("got", "go")]`

With a male voice (Alexandre) and a female voice (Mae) we recorded each of these sounds.

- Stimuli Pairing: We made 36 pairs of different sounds for each voice, and another 36 pairs of the same sounds for each voice, which brings the total to 144 pairs.
- Nasal vs. Oral: Of the 72 pairs, half did not contain a nasal sound (focusing on oral vowels such as "bat", "bet", "bot") for later statistical significance studies.
- Technical Processing: We developed software to help us record the sounds quickly and consistently. The software sets the same recording specifications for both of us: `samplerate=44100`, `channels=1`. It also trims the audio to the same length (removing silences at the beginning and end) and reduces background noise as much as possible.
- Pair Generation: We used a Python script to group stimuli by their consonantal structure (onset and coda) to ensure the pairs generated were phonologically relevant for testing vowel perception.

Procedure

The procedure was designed to be user-friendly, statistically robust, and above all accessible. We therefore developed a website (https://prosody-exp.duckdns.org/) on which to take the test and distributed the link to people willing to participate in the study. The welcome text reads : “ Vowel Perception Study Auditory Perception Research Vowel Perception Study You will listen to pairs of spoken words and decide whether the two sounds are the same or different. The study takes approximately 10–15 minutes. Please use headphones in a quiet environment. “. The participants then fill a form providing insight on their French proficiency level.

- Instructions: The participant was instructed to listen to pairs of syllables and decide if they were the "same" or "different."
- Equipment: We recorded whether the participant used headphones (25) or speakers (11)
- Randomization: The trial presentation is randomized to avoid order bias.
- Continuity: We ensured that participants would not have to start over if they returned later, as this would be equivalent to untracked training that could skew the results. The task takes approximately 5–10 minutes to complete. No forced pauses were included and there was no way to take a break.

Results

Summary of hypotheses and their results:

Main hypothesis :

H0: Perception of French vowels is the same for native and non-native speakers.

Sub-hypothesis:

H0: Accuracy is the same for Native and Non-Native people.

P-val=0.1509316314311325

P-corrected= 0.301863262862266

=> Rejected

H0: Reaction time is the same for Native and Non-Native people

P-val=0.05251534164566762

P-corrected= 0.1050306832913352

=> Rejected

Robustness proving hypothesis:

H0: Perception of French vowels is the same for native and non-native speakers, independently of other modalities or factors.

=> Rejected

Sub-hypothesis:

H0: Accuracy is the same for native and non-native speakers, independently of other modalities or factors.

=> Rejected

**Here, group 1 is all the 1st sub modalites: headphones, low French level, low French listening rate, few years of French practice, feminine gender, and low reaction time.*

group 2 is the 2nd sub modality: speaker, high French level, high French listening rate, more years of French practice, male,default gender, and high reaction time.

Global is the base group with all the native by one side and all the non-native by the other

Sub-sub-hypothesis:

H0: Accuracy difference is the same between group 1 and global

p-val=0.0732036362436237

p_corrected=0.21961090873087108

=> Rejected

H0: Accuracy difference is the same for group 1 and 2

p-val=0.640625

p_corrected=1.0

=> Rejected

H0: Accuracy difference is the same for group 2 and global

p-val=0.7421875

p_corrected=1.0

=> Rejected

Sub-hypothesis:

H0: Reaction time is the same for native and non-native speakers, independently of other modalities or factors.

=> Rejected

Sub-sub hypothesis

H0: Reaction Time difference is the same for group 1 and global

p-val=0.0597328978159432

p-corrected=0.1791986934478296

=> Rejected

H0: Reaction Time difference is the same for group 1 and 2

p-val=0.5168653215825211

p-corrected=1.0

=> Rejected

H0: Reaction Time difference is the same for group 2 and global

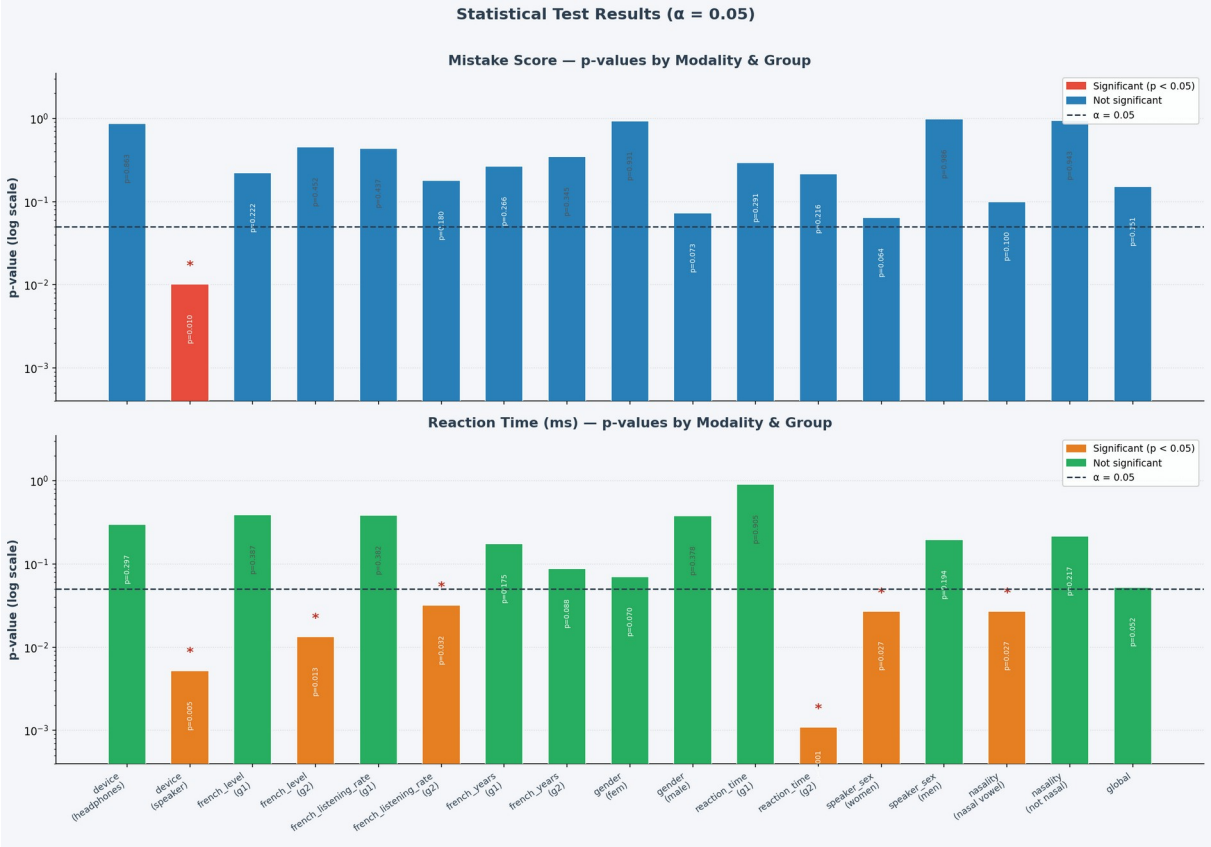
p-val=0.0157414657506571

p-corrected=0.04722439725197129

=> Accepted

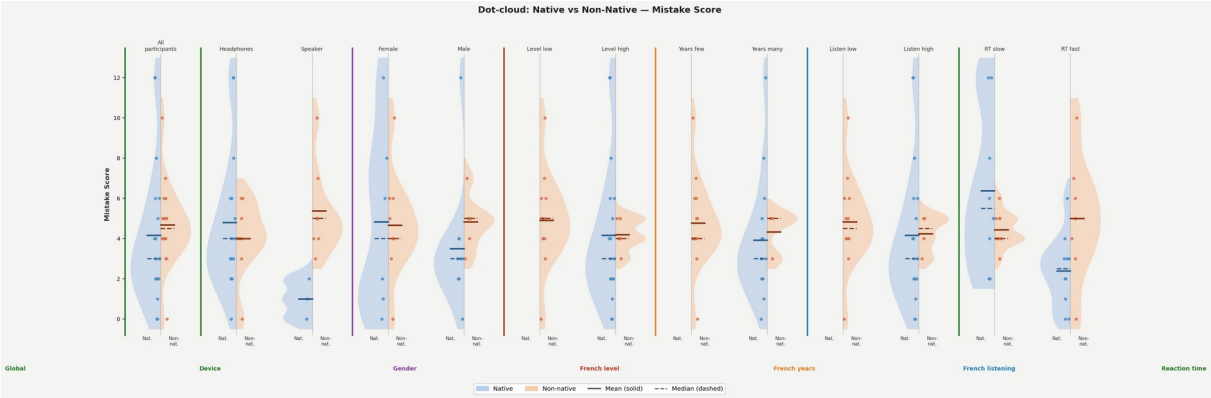
Report Prosody (S. Ruvoletto)

Each modalities p-value:

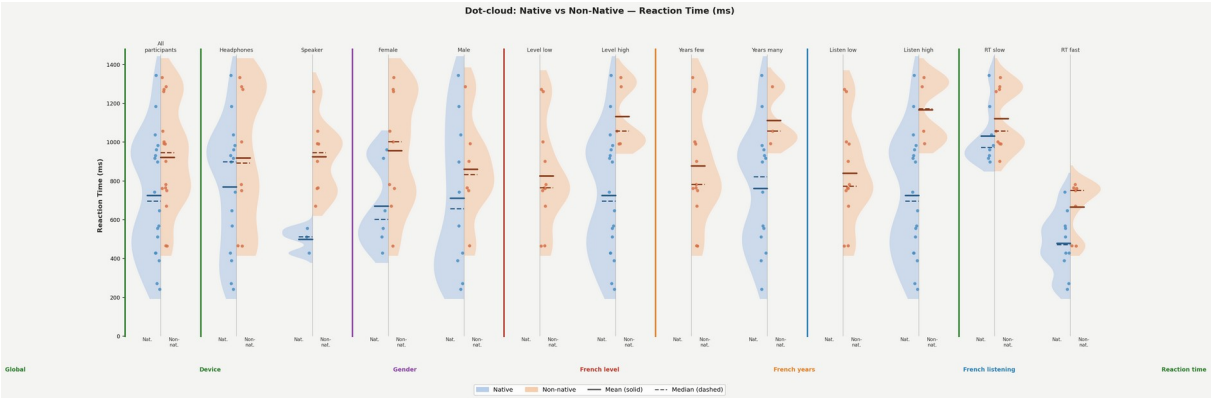


Statistical informations graph:

Accuracy:



Reaction Time:



Discussion and Conclusions

This work was carried out in an academic context with constraints of time and resources. The results are heavily influenced by the size of the dataset and by certain approximations in both the experimental design and the statistical processing (grouping choices, choice of modalities).

This must be kept in mind when interpreting the results.

Regarding the results, native and non-native speakers of French appear to differ in their perception of French phonemes, showing different reaction times and different accuracy when discriminating one phoneme from another.

However, this finding is open to criticism: depending on the sub-group, the differences in both accuracy and reaction time between the two groups are not consistent from a statistical standpoint.

Therefore, while the nativity of a speaker can be a strong indicator of their perceptual capacity in the studied language, perception is also highly dependent on many factors that may allow a non-native speaker to reach the same perceptual abilities as a native speaker.